



Tribhuvan University
Faculty of Humanities and Social Sciences
Bachelor of Computer Application (BCA)

Course Code: BCA 103
Title: Digital Logic

Full Marks: 60
Pass Marks: 24
Time: 3 Hours

*Candidates are required to answer the questions in their own words as far as practicable.
Figures in the margin indicate full marks. Attempt the questions as instructed.*

Group A
(Very Short Questions)

1. Attempt all questions. (10*1=10)
- Define analog signal.
 - What is a binary number system?
 - What is the need for complement?
 - Define a Gray code. Find gray code of 1011010
 - What is a Boolean algebra? How is it different from Binary System?
 - What is a basic logic gate?
 - Define a combinational circuit.
 - What is the race around condition?
 - What is a register?
 - Define a counter.

Group B

Attempt any SIX questions (6 × 5 = 30 marks)

- Convert the following:
 - $(101101)_2$ to decimal
 - $(45)_{10}$ to binary
- Subtract 23 from 18 using both 1's and 2's complement method.
- Simplify Boolean expressions using K-Map
 $F(A,B,C) = \sum(0,2,4,6)$ and $d = \sum(1,3)$
- Describe the working of half adder with logic diagram.
- Explain multiplexer with applications.
- Explain bidirectional shift register.





8. Explain different types of flip-flops (RS, JK) with truth table.

Group C

Attempt any TWO questions ($2 \times 10 = 20$ marks)

9. Write the steps for designing a combinational circuit. Design a 4-bit magnitude comparator and explain its working principle.
10. A JK flip-flop is subjected to continuous clock pulses and both J and K inputs are HIGH.
 - a. Explain what happens in a simple JK flip-flop.
 - b. Explain how a Master-Slave configuration resolves this issue.
 - c. Illustrate your answer using a timing diagram
11. Design a 4-bit binary counter using JK flip-flops.
 - a. Show logic diagram
 - b. Explain counting sequence
 - c. Analyze timing behavior



Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN
Model Question 2026



Bachelor in Computer Applications
Course Title: Computer Fundamental and Application
Code No: BCA 101
Semester: 1st

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Candidates must answer the question in their own words as far as possible.

Group A (Brief Answer Questions)

Attempt All the Questions.

10x1=10

1. Define anatomy of Computer.
2. Write the main electronic component use in fourth generation of computer.
3. What are the main components of CPU?
4. What is system BUS?
5. Define language Antivirus.
6. What is programming language?
7. Define internet and intranet.
8. What do you mean by cryptography
9. Write the nay two application of AI.
10. Define Virtual and Augmented reality.

Group B (Short Answer Question).

Attempt any SIX Question only.

[6x5=30]

1. What is computer? explain any five characteristics of computer.
2. Write the different between primary and secondary memory with example.
3. What is motherboard? Write and explain main components of motherboard.
4. What is language translator? Write and explain its types.
5. What do you mean by database model? Explain its types.
6. Define OSI model and explain its layers.
7. What contemporary technology? explain blockchain and its application in information technology and other fields.

Group C (Long Answer Question)

Attempt any TWO Questions only.

[2X10=20]

1. What is Computer Architecture? Explain its main components with diagram.
2. Define operating system and explain its function
3. What is computer security? Why it is important in Information technology? Explain the types of cryptography with suitable example.





Faculty of Humanities and Social Science

Office of the Dean

2026

(Model Question for CAMT 104)

Bachelor In Computer Applications

Course Title: Mathematics I

Code No: CAMT 104

semester 1st

Full Mark: 60

Pass Marks:24

Time: 3 hours

Candidates are required to answer the questions in their own words as far as possible.

Group A

Short answer questions

[10 x1=10]

Q.1) Write contrapositive of the statement "If you are I.T. student then you read mathematics".

Q.2) Find absolute value of the complex number $\frac{3-4i}{3+4i}$.

Q.3) What is the sum of the series $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16}$.

Q.4) If A is a matrix of size $m \times n$ and B is a matrix of size $p \times q$ and multiplication BA is defined, then find the size of the matrix BA.

Q.5) Write equivalent Cartesian equation of the polar equation $r \sin \theta = 2$.

Q.6) Write foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$.

Q. 7) For any two non-parallel vectors $\vec{a}$ and $\vec{b}$ write unit vectors perpendicular to both $\vec{a}$ and $\vec{b}$.

Q. 8) Write vector area of the parallelogram having two adjacent sides $\vec{a}$ and $\vec{b}$.

Q. 9) Write permutation of 12 numbers in a clock.

Q. 10) How many ways are there to arrange the word "PETROL", which do not begin with P?

Group B

Attempt any six questions

[6x5=30]

Q. 11) Solve the inequality $3 + 2x - x^2 \geq 0$.



Q.12) If a, b, c , and d are in G.P. prove that $a^2 - b^2, b^2 - c^2, c^2 - d^2$ are also in G.P.

Q.13) Prove that $\begin{bmatrix} 1+x & 1 & 1 \\ 1 & 1+y & 1 \\ 1 & 1 & 1+z \end{bmatrix} = xyz \left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z} + 1 \right)$

Q.14) Find the equation of the ellipse whose latus rectum is 5 and the eccentricity is $\frac{1}{\sqrt{2}}$.

Q.15) If $\vec{a} \cdot \vec{b} = \sqrt{3}$ and $\vec{a} \times \vec{b} = (1, 2, 2)$ find the angle between $\vec{a}$ and $\vec{b}$.

Q.16) How many numbers of three different digits less than 500 can be formed from the integers 1, 2, 3, 4, 5, and 6?

Q.17) Find the domain and range of the function $f(x) = \sqrt{9 - x^2}$.

Group C

Attempt any two questions

[10x2=20]

Q.18) a. Find Eigen values and corresponding Eigen vector of the matrix $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$

b) Let $A = \begin{pmatrix} 1 & -1 \\ 3 & 5 \\ 2 & 7 \end{pmatrix}$, $u = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ define a transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$, by $T(x) = Ax$ find $T(u)$.

Q. 19) a. Find sum of squares of the first n natural numbers.

b. Find a vector and unit vector perpendicular to the plane of $P(1, -1, 0)$, $Q(2, 1, -1)$ and $R(-1, 1, 2)$.

Q. 20) a. Find the equation of a hyperbola in standard form having focus $(-2, 0)$ and Diretrix $x = -\frac{1}{2}$.

b. In an examination paper on mathematics, 20 questions are set. In how many different ways you can choose 18 questions to answer?

The End



Tribhuvan University
Faculty of Humanities and Social Sciences
Bachelor in Computer Applications (BCA)
2026

Course Code: (BCA105)
Title: Professional Communication and Ethics

Full Marks: 60
Pass Marks: 30
Time: 3 Hours

Candidates are required to answer all the questions in their own words as far as practicable. Figures in brackets indicate full marks.

Group 'A'
(Very Short Questions)

1. Answer all questions.

(10 x 1 = 10)

- A. What do you mean by an algorithm?
- B. In the context of non-verbal communication, what does the term *Proxemics* refer to?
- C. Which specific part of a letter represents the "short, polite closing?"
- D. What is the "Stone Soup Effect" a metaphor for in group communication?
- E. In "The Thief's Story," what was the name the narrator used to stay ahead of the police?
- F. What is the difference between a heading and a letterhead in professional correspondence?
- G. According to the text on emotional intelligence, which is often considered more important for professional success: IQ or EQ?
- H. Define whistle-blowing as it relates to IT professional integrity.
- I. What does the term 'concord' refer to?
- J. What does the author mean by 'Dimmer dilemma' in Garber's "Death by PowerPoint?"

Group 'B'
(Short Answer Questions)

Answer any SIX Questions.

(6 x 5 = 30)

2. Describe the "Talk, Don't Read" principle of presentation skills.
3. What are the key differences between formal and informal letters? Mention at least three structural differences.
4. Define sociogram and how it helps in understanding team dynamics.
5. Discuss the importance of active listening and its three main degrees (Repeating, Paraphrasing, and Reflecting).
6. Based on the reading "The Digital Citizen," explain how digital footprints affect a professional's reputation.
7. Rewrite the following sentences using the verbs in the brackets in the correct tense.
 - a) My father (not drink) coffee very often. He (drink) it now.





- b) My sister (work) in an American company at the moment.
 - c) Look at the Bagmati River. It (flow) faster than usual.
 - d) Normally I (go) to bed at 11 am.
 - e) I (hear) some strange noise now.
8. Fill in the blanks with at, on and in.
- a) The old temple is hidden deep the forest.
 - b) We waited the bus stop for over an hour.
 - c) The birds were flying the sky in perfect formation.
 - d) She placed the vase the table.
 - e) He arrived the airport just in time for his flight.

Group 'C'
(Long Answer Questions)

Answer any TWO Questions:

(2 x 10 = 20)

9. You recently bought a micro-oven from an online retailer, paying 10500 rupees. After using about one week, the case of the oven started heating excessively and consumed more power. At the same time, your brother found a similar oven for 8000 rupees in a local shop. Write a complaint letter to the online retailer, supply.com, Pepsikola, Kathmandu.
10. Define Netiquette. Discuss the ethical implications of "Flame Wars" and "Cyberbullying" in professional cyberspace. How can practicing 'The Human Element' and 'Compassion' improve digital interactions?
11. How did the guide demonstrate leadership during the refugees' journey, and what qualities made his leadership effective in a life-threatening situation in "The Great Answer?"

.....**The End**.....





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Course Code: BCA 102
Title: Programming in C

Full Marks: 60
Pass Marks: 24
Time: 3 Hours

*Candidates are required to answer the questions in their own words as far as practicable.
Figures in the margin indicate full marks. Attempt the questions as instructed.*

Group A
(Very Short Questions)

1. Attempt all questions.

(10*1=10)

- What are keywords and identifiers ?
- Define recursion.
- What is call by value?
- Write syntax for declaring an array.
- What is a structure?
- Define pointer.
- What is dynamic memory allocation?
- What is a file in C?
- What is command-line argument?
- Name any one graphics function in C.

Group B

Attempt any SIX questions (6 × 5 = 30 marks)

- Explain with example how a c program is compiled and executed.
- Differentiate between call by value and call by reference with suitable example.
- Write a program to print the following pattern using loop:

```
*  
*      *  
  
*      *      *  
*      *      *      *
```



Programming in C



5. Describe any 3 string handling functions in C with example.
6. Explain structures in C. How are arrays of structures used?
7. Explain how malloc(), calloc() and free() are used to allocate memory dynamically.
8. Write a program to draw a circle at the center of the screen.

Group C

Attempt any TWO questions ($2 \times 10 = 20$ marks)

9. Write a C program to perform matrix multiplication using two-dimensional arrays. Explain how arrays and functions are used together in C.
10. Explain nested structures with example. Design a structure for student records (name, roll, marks) and write a program to store and display data for 5 students.
11. Write a program to create a file, write data, and read it back. Explain file handling functions used. Discuss random file access (fseek, ftell) with example.

